Practical No.: 6

Title: Create a Single Page Application (SPA) using ReactJS

Aim:

To develop a **Single Page Application (SPA)** in ReactJS that dynamically loads different content sections without refreshing the page.

Theory

A **Single Page Application (SPA)** is a web application that loads a **single HTML page** and dynamically updates the content using JavaScript, instead of loading new pages from the server. ReactJS is widely used for developing **SPAs** because of its component-based architecture and efficient re-rendering using the **Virtual DOM**.

Concepts Covered:

- **React Router**: Used to navigate between different pages/components without a full page reload.
- **SPA Architecture**: Rendering different views dynamically based on the URL.

Syntax for Routing in ReactJS:

```
import { BrowserRouter as Router, Route, Routes, Link } from "react-router-dom";
```

.ReactJS Code

Step 1: Create `index.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Practical 6: Single Page Application</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js"></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-
dom.development.js"></script>
    <script src="https://unpkg.com/babel-standalone@6/babel.min.js"></script>
</head>
<body>
    <h2>Practical 6: Single Page Application</h2>
    <div id="root"></div>
    <script type="text/babel" src="demo.js"></script>
</body>
</html>
```

Step 2: Create a JavaScript file (app.js)

```
const { BrowserRouter, Routes, Route, Link } = ReactRouterDOM;

// Home Component
function Home() {
    return <h2>Welcome to the Home Page</h2>;
}

// About Component
function About() {
    return <h2>This is the About Page</h2>;
}

// Contact Component
function Contact() {
    return <h2>Contact Us at: example@example.com</h2>;
}

// App Component with Navigation
function App() {
    return (
        <BrowserRouter>
            <div>
                <nav>
                    <ul>
                        <li><Link to="/">Home</Link></li>
                        <li><Link to="/about">About</Link></li>
                        <li><Link to="/contact">Contact</Link></li>
                    </ul>
                </nav>

                <Routes>
                    <Route path="/" element={<Home />} />
                    <Route path="/about" element={<About />} />
                    <Route path="/contact" element={<Contact />} />
                </Routes>
            </div>
        </BrowserRouter>
    );
}

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<App />);
```